

Sprint 3 Report

Nifty100 Analytics: Screener Engine, Peer Analytics, Excel Reports, and QA Validation

Report Date	2026-07-09
Repository	D:\bssprint1\Nifty100
Database	db/nifty100.db
Primary Outputs	output/screener_output.xlsx; output/peer_comparison.xlsx; reports/radar_charts; reports/Sprint3_Report.pdf

Validation Area	Result
ETL tests	PASS - 50 passed, 0 failed
Full project tests	PASS - 416 passed, 0 failed
Peer percentile table	PASS - 7,061 records available
Radar charts	PASS - 92 company radar PNG files available
Excel reports	PASS - screener workbook has 6 sheets; peer workbook has exactly 11 sheets

1. Sprint Goal

Sprint 3 focused on turning the Nifty100 financial data foundation into usable investment analytics outputs. The goal was to deliver a repeatable screener engine, peer comparison engine, percentile rankings, radar chart visuals, formatted Excel workbooks, and automated validation covering data quality and analytics behavior.

The sprint output is intended for analyst review: companies can be screened using preset strategies, compared within peer groups, ranked by composite quality, and exported into workbooks with conditional formatting and median benchmarks.

2. Architecture

The application uses SQLite as the analytical store and a Python analytics layer for ratios, scoring, peer comparison, report generation, and tests.

Layer	Component	Responsibility
Data store	db/nifty100.db	Stores companies, sectors, market data, financial statements, financial ratios, and peer percentiles.
KPI layer	src/kpi	Calculates profitability, leverage, CAGR, cash quality, and peer metrics.
Screener layer	src/screener	Loads financial data, applies filters and presets, computes rankings, and exports screener output.
Peer layer	src/analytics/peer.py and peer_report.py	Builds peer groups, computes percentile ranks, exports 11 peer comparison sheets, and applies benchmark formatting.
Outputs	output/ and reports/	Stores screener workbook, peer workbook, radar charts, and sprint documentation.
Tests	tests/	Validates ETL rules, ratios, CAGR logic, screener behavior, peer analytics, and report generation.

3. Screener Engine

The screener engine loads company financial data, enriches it with calculated ratios, computes composite quality signals, applies filter rules, and exports strategy-specific results. It supports both raw filter execution and preset-based screening.

- Core input tables include companies, sectors, financial_ratios, profitandloss, balancesheet, cashflow, and market_cap.
- The engine derives measures such as debt/equity, asset turnover, interest coverage, CAGR fields, CFO quality, and composite quality score.
- The final screener output is written to output/screener_output.xlsx with six strategy sheets.

4. 15 Filters

The screener supports the main analyst filters needed to shortlist quality, growth, valuation, balance sheet strength, income, and operating performance. The Sprint 3 report tracks the following 15 filter families:

No.	Filter	Purpose
1	ROE	Measures shareholder return efficiency.
2	ROCE	Measures return generated on capital employed.
3	Operating Margin	Screens operating profitability.
4	Net Profit Margin	Screens bottom-line profitability.
5	Debt Equity	Controls financial leverage.
6	Interest Coverage Ratio	Checks ability to service debt.
7	Free Cash Flow	Screens cash generation.
8	Revenue	Filters company scale by sales.
9	Net Profit / PAT	Filters earnings scale.
10	Market Cap	Controls company size and liquidity.
11	Dividend Yield	Identifies income-oriented companies.
12	Revenue CAGR	Measures top-line growth.
13	PAT CAGR	Measures earnings growth.
14	EPS CAGR	Measures per-share growth.
15	Asset Turnover	Screens asset utilization efficiency.

5. Six Presets

The Excel screener output contains six strategy presets. Each preset uses the common financial base but emphasizes a different analyst workflow.

Preset	Intent
Quality Compounder	High-quality companies with strong ROE and controlled leverage.
Value Pick	Companies that pass valuation-oriented and quality constraints.
Growth Accelerator	Companies with stronger revenue, PAT, and EPS growth momentum.
Dividend Champion	Income candidates with dividend yield and financial discipline.
Debt-Free Bluechip	Large, high-quality companies with very low or zero leverage.
Turnaround Watch	Candidates that require closer review due to improving or unusual signals.

6. Composite Score

The composite score converts multiple financial dimensions into a comparable 0-100 quality signal. It is used by the screener output and peer report benchmark logic.

Score Block	Weight	Included Signals
Profitability	35	ROE, ROCE, operating margin
Cash Quality	30	Free cash flow, CFO/PAT quality, positive cash generation
Growth	20	Revenue CAGR and PAT CAGR
Leverage	15	Debt discipline and interest coverage

Score Block	Weight	Included Signals
Total	100	Composite quality score and related rank fields

The scoring engine also supports sector-relative comparison, helping avoid unfair comparisons between business models with structurally different margins, leverage, or capital intensity.

7. Peer Engine

The peer engine groups companies by comparable operating context, computes metric percentiles, and produces a peer workbook for analyst review. The peer_percentiles database table currently contains 7,061 records.

- Percentile metrics include ROE, ROCE, net profit margin, debt, ICR, FCF, EPS, asset turnover, dividend payout, and total debt.
- For low-is-good metrics such as debt, percentile calculation is inverted so lower debt receives a stronger percentile.
- The highest composite quality score in each peer sheet is treated as the benchmark company and highlighted in amber/gold.
- Each peer sheet includes a Peer Median row at the bottom for quick group-level comparison.

8. Radar Charts

Radar charts are generated under reports/radar_charts. The directory currently contains 92 company radar PNG files. These charts visualize company performance against peer or market benchmarks across key quality and risk dimensions.

Artifact	Status
reports/radar_charts/*_radar.png	PASS - 92 PNG files available
Company-level radar visuals	Available for analyst review and dashboard integration

9. Excel Reports

Sprint 3 produces two primary Excel workbooks: a screener workbook for strategy output and a peer comparison workbook for sector and peer analysis.

Workbook	Sheets	Validation
output/screener_output.xlsx	6 strategy sheets	PASS - Quality Compounder, Value Pick, Growth Accelerator, Dividend Champion, Debt-Free Bluechip, Turnaround Watch
output/peer_comparison.xlsx	11 peer sheets	PASS - IT Services, Banking, FMCG, Automobile, Pharma, Energy, Metals, Telecom, Consumer, Finance, Others

The peer workbook includes required company identifiers, financial KPI columns, 10 percentile columns, conditional colors for percentile strength, benchmark row highlighting, and peer median rows. Percentile coloring uses green for 75 and above, yellow for 25 to 75, and red for 25 and below.

Color	Condition	Meaning
Green	Percentile >= 75	Top performer

Color	Condition	Meaning
Yellow	25 < Percentile < 75	Average performer
Red	Percentile <= 25	Weak performer
Gold / Amber	Highest composite_quality_score in sheet	Benchmark company

10. Testing Results

The project test suites were run after Sprint 3 implementation. The final status has no failures. The full run reports two existing NumPy runtime warnings during screener scoring tests, with all assertions passing.

Command		Result
python -m pytest tests/etl -v		50 passed, 0 failed
python -m pytest tests/ -v		416 passed, 0 failed, 2 warnings
Area	Status	
ETL Tests	PASS	
Ratio Tests	PASS	
CAGR Tests	PASS	
Screener Tests	PASS	
Peer Tests	PASS	

Additional manual validations completed: Quality Compounder top five companies satisfy ROE greater than 15 and debt/equity below 1; IT Services peer ranking confirms the highest ROE company also has the highest ROE percentile.

11. Conclusion

Sprint 3 deliverables are complete. The repository now has a working screener engine, six preset-driven screener sheets, a peer comparison workbook with exactly 11 sheets, peer percentile formatting, benchmark highlighting, peer median rows, radar chart outputs, and passing automated tests.

The analytics outputs are ready for stakeholder review and can serve as the base for Sprint 4 dashboard integration, improved visual summaries, or expanded investment workflow automation.